# MAT244 (ODEs) Notes

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# 1 Introduction to Ordinary Differential Equations

*This lecture corresponds to §1.1-1.3 in the textbook.*

## 1.1 What Are Differential Equations?

An **differential equation** (which I will often abreviate as a **DE**) is an equation containing derivatives of an unknown function. The two main types of DEs are **ordinary differential equations**, or **ODEs**, which are equations involving functions depending on a single independent variable, and **partial differential equations**, or **PDEs**, which are equations involving functions which depend on more than one independent variable. For example, the equation:

$$\frac{d^2u}{dt^2} + \frac{du}{dt} + u = f(t)$$

is an ODE that depends on the independent variable $t$. The equation:

$$-\partial_t^2 u + \partial_x^2 u + +\partial_y^2 u + \partial_z^2 u = 0$$

is an example of a PDE that depends on multiple independent variables $x$, $y$, $z$, and $t$. The focus of this course will be on ODEs.

**Example 1.1.** *Consider the following differential equation:*

$$\frac{dy}{dt} = -y(t) + t$$

We know that this is an ODE, but what does it represent? From calculus, we know that $dy/dt$ is the derivative of $y(t)$, and that this derivative represents how fast the function $y$ is changing with respect to $t$. Since we have that $dy/dt = -y(t) + t$, we can infer that $t$ will decrease as $y$ increases and will tend to increase as $t$ increases.

Recall that $\frac{dy}{dt}\big|_p$ represents the slope of $y(t)$ passing through the point $p$. We can thus say that the **direction** of $y(t)$ at point $t = t_0$ is equal to $-y(t_0) + t_0$. With this information, we can plot a **direction field** (or **slope field**) for this differential equation:

**Example 1.2.** *Consider the following differential equation:*

$$\frac{dy}{dt} = ay$$

*where $a \in \mathbb{R}$. Plot direction fields for $a = -2, -1, 0, 1, 2$.*

## 1.2 Solutions of Differential Equations

We "solve" differential equations of the form $dy/dt = f(t, y)$ by finding a function $y(t)$ that satisfies it.

**Example 1.3.** *Solve $y' = 2y(t)$*

*Solution.* First, rewrite the equation so that all the terms with $y$ appear on one side of the equation and all the terms with $t$ appear on the other:

$$\frac{dy}{dt} = 2y \implies \frac{dy}{y} = 2dt$$

Now we integrate both sides:

$$\int \frac{dy}{y} = \int 2dt$$
$$\ln|y| = 2t + C$$

Now solve for $y$:

$$\ln|y| = 2t + C$$
$$|y| = e^{2t+C}$$
$$y(t) = \pm e^{2t} e^{C}$$

Note that $e^C$ is a positive constant, so we can replace it with a new constant $A = e^C$ such that:

$$y(t) = Ae^{2t}$$

Lastly let us check our work by integrating $y(t)$ and ensuring it matches the given equation:

$$y' = 2Ae^{2t}$$
$$y' = 2y(t)\checkmark$$

Suppose that we are also given an initial condition, for instance suppose for the previous example that $y(0) = 1$. This initial condition uniquely determines the value of the constant and hence the function $y$. With the above example, if we assume $y(0) = 1$, then we find that:

$$y(0) = 1 = Ae^{2(0)} \tag{1.1}$$
$$1 = A \tag{1.2}$$

**Example 1.4.** *Solve* $\frac{dy}{dx} = 3y - 2$*, with the initial value* $y(0) = 2$.

*Solution.* First put all the terms with $y$ on one side and the terms with $x$ on the other:

$$\frac{1}{3y-2}dy = dx$$

Now integrate both sides:

$$\int \frac{1}{3y-2}dy = \int dx$$
$$\frac{1}{3}\ln|3y-2| + C_1 = x + C_2$$

If we put the constants together we obtain:

$$\begin{aligned}
\ln|3y-2| &= 3x + C \\
|3y-2| &= Ae^{3x} \\
3y - 2 &= \pm Ae^{3x} \\
3y &= \pm Ae^{3x} + 2 \\
y &= \frac{\pm Ae^{3x}+2}{3}
\end{aligned}$$

Now solving for the initial value:

$$\begin{aligned}
y(0) = 2 &= \frac{\pm Ae^{3(0)}+2}{3} \\
2 &= \frac{A+2}{3} \\
6 &= A + 2 \implies A = 4
\end{aligned}$$

So a solution to the initial value problem is $y(t) = \frac{4e^{3t}+2}{3}$.

## 1.3 Classification of Differential Equations

The **order** of a DE is the highest derivative in the equation. For example, the order of $y' = y(t)$ is 1, and the order of $y'' + y + t = y(t)$ is 2. The general form of an order $n$ DE is:

$$F(x, u(x), u'(x), u''(x), \ldots, u^{(n)}(x)) = 0$$

A solution of the ODE $y^{(n)} = f(x, y, y', \ldots, y^{(n-1)})$ on the interval $a < x < b$ is a function $\phi$ such that $\phi', \phi'', \ldots, \phi^{(n)}$ exists and satisfies $\phi^{(n)} = f(x, \phi, \phi', \ldots, \phi^{(n-1)})$ for all $x \in (a, b)$. Note that $\phi$ and all necessary derivatives must exist on $a < x < b$ and $\phi$ must satisfy the differential equation.

For example, consider $y'' + y = 0$, or $y'' = -y$. Solutions to this DE will be of the form $e^{ix}$ or make use of sines and cosines (for instance, it is easy to check that $y = \sin(x) + \cos(x)$ is a solution). However, naturally the question arises: does a solution always exist, if so it is unique? And how do we find it? We will learn how to answer those questions in the next lecture.